

DATA SCIENCE AND PYTHON COURSE

Capstone Project

Customer Churn Prediction — One-Day Build

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Project Overview

This capstone brings together everything you have learned so far — Python fundamentals, statistics, NumPy, Pandas, data visualization, machine learning, and feature engineering — into one complete, resume-ready project. You will build a model that predicts whether a telecom customer is likely to leave (churn), using the Telco Customer Churn dataset.

- **Dataset:** Telco Customer Churn (Kaggle)
- **Problem type:** Binary Classification (Churn: Yes / No)
- **Duration:** 4–6 hours
- **Deliverable:** One clean Jupyter Notebook with explanations, code, and a short business summary

Session Schedule

Time Block	Focus	What You'll Do
Hour 1	Setup & EDA	Load data, check shape/info/describe, find missing values, plot churn count, tenure histogram, and charges boxplot. Discuss correlation vs causation.
Hour 2	Cleaning & Feature Engineering	Fix TotalCharges data type, handle missing values, encode categorical columns, scale numeric columns with StandardScaler, drop customerID, and split train/test (80/20).
Hour 3	Model Building	Train a Logistic Regression baseline and a Decision Tree Classifier. Optional bonus: run K-Means on tenure and charges to explore customer segments.
Hour 4	Evaluation & Wrap-Up	Compare models using accuracy, precision, recall, and confusion matrix. Decide which metric matters most for churn. Write 3–4 business insight bullets.

Skills You Are Applying

Every step of this project connects directly to a topic from the course:

- **Python & Statistics:** Pandas for loading and exploring data; descriptive statistics and spread of data for summarizing churn patterns.
- **Visualization:** Matplotlib and Seaborn charts to understand distributions and relationships.
- **Feature Engineering:** Encoding categorical data and scaling numeric features with StandardScaler.
- **Machine Learning:** Train/test split, Logistic Regression, Decision Tree Classifier, and model evaluation metrics.
- **Bonus:** K-Means Clustering to view customer segments beyond the prediction task.

What to Submit

Submit a single, well-organized Jupyter Notebook containing:

- Clear markdown headers for each of the four session sections
- Code cells with short explanatory comments
- Charts that support your findings
- A 4–5 line summary at the end covering: the problem, your approach, the best-performing model, and one key business insight

Key Reminder

Think like a business analyst, not just a coder.

A good accuracy score is not the finish line. Ask yourself: which mistake costs the company more — telling a loyal customer they might leave, or missing a customer who actually leaves? This is what interviewers want to hear you explain.

Good luck — build something you're proud to show in an interview!